Atiyah-Hirzebruch's Counterexample to the Integral Hodge Conjecture

By

Baran Cetin, Yijie Pan

University of Southern California

Los Angeles, California

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1 Some Notes on Topology

This section will be some notes on topology that are closely related to the construction of the counterexample.

1.1 Stable Cohomological Operations

Let **Ab** be the category of Abelian groups, and **CW** be the category of CW complexes. If n is an integer and G is an Abelian group, then $H^n(-;G) : \mathbf{CW} \to \mathbf{Ab}$ is a contravariant functor.

Definition 1.1 (Stable Cohomological Operations)**.** A cohomological operation of type (n, q, π, G) is a natural transformation from $H^n(-;\pi) \to H^q(-;G)$. denoted as $\mathcal{O}(n, q, \pi, G)$. In particular, fix an integer k, a stable natural transformation of degree k is a family of natural transformations $\{\varphi^i \in \mathcal{O}(i, i+k, \pi, G)\}_{i\in\mathbb{Z}}$ so that $\varphi^n_X \cong \varphi^{n+1}_{\Sigma X}$ via the suspension isomorphism for any integer n, where ΣX is the suspension of X.

Theorem 1.1 (Classfication of Cohomological Operations)**.** *There exists a bijection* $\mathcal{O}(n, q, \pi, G) \cong H^q(K(\pi, n), G)$.

Proof. By Brown's representability theorem, there exists a natural isomorphism $H^n(-;\pi) \cong$

$[-, K(\pi, n)] = \mathrm{Hom}(-, K(\pi, n))$ in the homotopy category. Apply Yoneda's lemma, one obtains a bijection $\mathcal{O}(n, q, \pi, G) \cong H^q(K(\pi, n), G)$. $\square$

One way to construct natural transformation is through the coefficient homomorphisms. Consider the exact sequence of coefficients $0 \to \mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}_p \to 0$ and $0 \to \mathbb{Z}_p \to \mathbb{Z}_{p^2} \to \mathbb{Z}_p \to 0$. One can directly verify that the functor $C^k(X; -)$ is an exact functor so that one obtains the following long exact sequence of cohomology groups:

$$\begin{tikzcd}
 & & & & H^{k+1}(X;\mathbb{Z}_p) \\
H^k(X;\mathbb{Z}) \arrow[r, "p"] & H^k(X;\mathbb{Z}) \arrow[r, "\sigma"] & H^k(X;\mathbb{Z}_p) \arrow[urr, "\beta"] \arrow[r, "\tilde{\beta}"] & H^{k+1}(X;\mathbb{Z}) \arrow[u, "\sigma"] & \\
H^k(X;\mathbb{Z}_p) \arrow[r] & H^k(X;\mathbb{Z}_{p^2}) \arrow[ur] & & &
\end{tikzcd} \tag{1}$$

The connecting homomorphisms β and $\tilde{\beta}$ are called the Bockstein homomorphims. The triangle in the diagram is commutative by the natural maps from the first cochain complexes to the second one.

Proposition 1.2. *Bockstein homomorphisms are cohomological operations.*

Proof. Because any map $f : X \to Y$ induces a homomorphism on chain complexes, it also commutes with the connecting homomorphism. $\square$

This motivates the definition of a stable cohomological operations called the Steenrod squares $Sq^i : H^k(-;\mathbb{Z}_2) \to H^{k+i}(-;\mathbb{Z}_2)$ which satisfies the following properties:

1. $Sq^0 = 1$; $Sq^1 = \beta$; $Sq^k(a) = a^2$; $Sq^n = 0$ if $n \geq k$;
2. $Sq^n(a \smile b) = \sum_{i+j=n} Sq^i(a) \smile Sq^j(b)$ for any n.

For p an odd prime, there exists a stable cohomological operations called the Steenrod powers $P^i : H^k(-;\mathbb{Z}_p) \to H^{k+2i(p-1)}(-;\mathbb{Z}_p)$ which satisfies the following properties:

1. $P^i(a) = a^p$ if $|a| = 2i$;
2. $P^n(a \smile b) = \sum_{i+j=n} P^i(a) \smile P^j(b)$ for any n.

Given the existence of the above cohomological operations, the composition

$$\mathcal{P}^1 : H^k(X;\mathbb{Z}) \xrightarrow{\sigma} H^k(X;\mathbb{Z}_p) \xrightarrow{P^1} H^{k+2(p-1)}(X;\mathbb{Z}_p) \xrightarrow{\tilde{\beta}} H^{k+2p-1}(X;\mathbb{Z}) \tag{2}$$

is a stable cohomological operation of degree $2p-1$. The next business in order is to show $\mathcal{P}^1 \neq 0$ and it suffices to give an example of space with an element $y \in H^*(X;\mathbb{Z})$

so that $\mathcal{P}^1(y) \neq 0$.

The space that plays an important role is $K(\mathbb{Z}_p, 1)$ for an odd prime p and the corresponding cohomology ring $H^*(\mathbb{Z}_p; H) := H^*(K(\mathbb{Z}_p, 1); H)$.

Proposition 1.3. *The infinite lens space $S^\infty/\mathbb{Z}_p$ is a $K(\mathbb{Z}_p, 1)$.*

Proof. The action of $\mathbb{Z}_p$ on S^∞ is a covering group action and S^∞ is a simply connected space. So $\pi_1(S^\infty/\mathbb{Z}_p) \cong \mathbb{Z}_p$. By the fibration exact sequence for homotopy groups, $\pi_k(S^\infty/\mathbb{Z}_p) = 0$ for $k > 1$. □

Proposition 1.4. $H^n(\mathbb{Z}_p, \mathbb{Z}_p) = \mathbb{Z}_p$ *if* $n \geq 0$.

Sketch of Proof. Use the cell structure. □

Proposition 1.5. $H^*(\mathbb{Z}_p, \mathbb{Z}_p) \cong \Lambda_{\mathbb{Z}_p}[u] \otimes \mathbb{Z}_p[v]$ *as a ring, where* $|u| = 1, |v| = 2$. *Moreover,* $v = \beta(u)$ *and* $\beta(v) = 0$, *where* $\beta : H^1(\mathbb{Z}_p, \mathbb{Z}_p) \to H^2(\mathbb{Z}_p, \mathbb{Z}_p)$ *is the Bockstein homomorphism;* $P^1(u) = 0$ *and* $P^1(v) = v^k$ *where* P^1 *is the first Steenrod Power.*

Corollary 1.6. *Let* $G = \mathbb{Z}_p \times \mathbb{Z}_p \times \mathbb{Z}_p$, $H^*(\mathbb{Z}_p, \mathbb{Z}_p) = \Lambda_{\mathbb{Z}_p}[u_1, u_2, u_3] \otimes \mathbb{Z}_p[v_1, v_2, v_3]$, *where* $v_i = \beta(u_i), P^1(u_i) = 0, P^1(v_i) = v_i^p$.

Proof. Since $K(G,1) = K(\mathbb{Z}_p,1) \times K(\mathbb{Z}_p,1) \times K(\mathbb{Z}_p,1)$, by the Kunneth formula for fields,

$$\begin{aligned} H^*(G,\mathbb{Z}_p) \cong & H^*(\mathbb{Z}_p,\mathbb{Z}_p) \otimes H^*(\mathbb{Z}_p,\mathbb{Z}_p) \otimes H^*(\mathbb{Z}_p,\mathbb{Z}_p) \\ = & \bigotimes_{i=1}^{3} \Lambda_{\mathbb{Z}_p}[u_i] \otimes \mathbb{Z}_p[v_i] \\ = & \Lambda_{\mathbb{Z}_p}[u_1,u_2,u_3] \otimes \mathbb{Z}_p[v_1,v_2,v_3]. \end{aligned}$$

Everything about Bockstein homomorphism and Steenrod square still holds in this ring. □

Theorem 1.7. *There exists an element $y \in H^4(G;\mathbb{Z})$ so that $\mathcal{P}^1(y) \neq 0$.*

Proof. Consider diagram (1). For any $y \in H^k(G;\mathbb{Z})$ with $k \geq 1$, $py = 0$, so $\mathrm{Im}(p) = 0 = \mathrm{Ker}(\sigma)$, so σ is injective and $\sigma(H^k(X;\mathbb{Z})) = \mathrm{Ker}(\tilde{\beta}) = \mathrm{Ker}(\beta)$, where the last equality is again by injectivity of σ.

Now consider $\beta(u_1u_2u_3) \in H^4(G;\mathbb{Z}_p)$, then $\beta(u_1u_2u_3) \in \mathrm{Ker}(\beta)$. But $\beta P^1(\beta(u_1u_2u_3)) = \beta P^1(v_1u_2u_3 - u_1v_2u_3 + u_1u_2v_3) = \beta(v_1^pu_2u_3 - u_1v_2^pu_3 + u_1u_2v_3^p) = \sum(-v_1^pv_2u_3 + v_1^pu_2v_3) \neq 0$. $\mathcal{P}^1(\sigma^{-1}(\beta(u_1u_2u_3))) = \sigma(\beta P^1(\beta(u_1u_2u_3))) \neq 0$ because σ is injective. □

This turns out to be the main topological obstruction to the integral hodge conjecture.

1.2 AHSS and d_3

(This section is only needed in the construction of the examples for Stein manifolds, not sure if I should write more about it.)

Theorem 1.8. *$d_3 \neq 0$ in the spectral sequence of $\mathbb{R}P^2 \times \mathbb{R}P^4$.*

Corollary 1.9. *Sq^3 is the only non-trivial stable cohomology operation from $H^n(-;\mathbb{Z}) \to H^{n+3}(-;\mathbb{Z})$.*

Corollary 1.10. *In Atiyah-Hirzebruch spectral sequence for K-theory, $d_3 = Sq^3$.*

2 Classic Cycle Map

2.1 K-theory and Cohomology for CW-complex

Definition 2.1. Let (X, Y) be a topological pair. We denote $\mathcal{K}(X, Y)$ as the set of functors ζ which satisfies the following conditions:

i: For any pair of finite simplicial complex (A, B) and any map $f : (A, B) \longrightarrow (X, Y)$, $f^!\zeta \in K^*(A, B)$.

ii: $f^!\zeta$ only depends on the homotopy class of f and if

$$\begin{tikzcd} (A, B) \arrow[r, "f"] & (X, Y) \\ (A_1, B_1) \arrow[u, "g"] \arrow[ur, "f_1"] & \end{tikzcd}$$

is a commutative diagram, then $g^!(f^!\zeta) = f_1^!\zeta$.

It follows from definition that $\mathcal{K}^*(X, Y) \simeq K^*(X, Y)$ is the usual K group for any finite CW-complex (X, Y). On the other hand, for a general CW-complex (X, Y), it is true that $\mathcal{K}^*(X, Y) \simeq \varprojlim_\alpha K^*(X_\alpha, Y_\alpha)$ where X_α runs through all the finite subcomplexes of X and $Y_\alpha = Y \cap X_\alpha$.

Similarly we can introduce the cohomology groups $\mathcal{H}^n(X, Y) \simeq \varprojlim_\alpha H^n(X_\alpha, Y_\alpha)$.

2.2 Analytic Cycles

Let X be a complex manifold and Y an complex analytic subspace of (complex) codimension q. Let $Y_{sing} \subset Y$ be the set of all singularities of Y, then Y is a complex

analytic subspace of codimension $\geq q+1$. $Y - Y_{sing}$ is a codimension q complex submanifold of $X - Y_{sing}$ and is canonically oriented with the same orientation as X, so one can define its poincare dual $y'' \in H^{2q}(X - Y_{sing})$, and this induces $y' \in \mathcal{H}^{2q}(X - Y_{sing})$ via the natural map. In order to define an element $y \in \mathcal{H}^{2q}(X)$, one consider the pullback of the inclusion $i : X - Y_{sing} \to X$, $i^* : \mathcal{H}^{2q}(X) \to \mathcal{H}^{2q}(X - Y_{sing})$.

Lemma 2.1. *$i^* : \mathcal{H}^{2q}(X) \to \mathcal{H}^{2q}(X - Y_{sing})$ is an isomorphism of groups.*

Proof. Proof is topological, will write later. □

By the lemma, in the following diagram:

$$\begin{tikzcd}
H^{2q}(X) \arrow[r, "\alpha"] \arrow[d, "i^*"] & \mathcal{H}^{2q}(X) \arrow[d, "i^*"] \\
H^{2q}(X - Y_{sing}) \arrow[r, "\alpha'"] & \mathcal{H}^{2q}(X - Y_{sing})
\end{tikzcd}$$

The right vertical map is an isomorphism.

Definition 2.2. $y = (i^*)^{-1}(\alpha'(PD[Y - Y_{sing}])) \in \mathcal{H}^{2q}(X)$ is called the analytic cycle induced by the analytic subspace Y.

Remark. *Here's how y as a functor works: let $f : A \to X$, where A is a finite CW-*

complex. One can find g homotopic to f so that $g : A \to X - Y_{sing}$, i.e., g doesn't touch the singularities. Then $f^! y = g^* PD[Y - Y_{sing}] \in H^{2q}(A)$.

Definition 2.3. An element $y \in \mathcal{H}^{2q}(X)$ is complex analytic if there exists $n_1, ..., n_k \in \mathbb{Z}$, $Y_1, ..., Y_k$ complex analytic subspaces, such that $y = \sum_{i=1}^{k} n_i y_i$, where y_i's are the complex analytic cycles induced by Y_i's.

3 Counterexample of the Integral Hodge Conjecture

3.1 The Main Theorem

Theorem 3.1. *Let X be a complex manifold and y a complex analytic cycle. Then in the Atiyah-Hirzebruch spectral sequence, $d_r y = 0$ for all r.*

3.2 Construction of the Counterexample

3.2.1 Stein Manifolds

Assume any $y \in H^{2q}(X;\mathbb{Z})$ is complex analytic, by corollary 1.10 and theorem 3.1, $\tilde{Sq}^3 y = 0$. But by theorem 1.8, $\mathbb{R}P^2 \times \mathbb{R}P^4$ has a nonvanishing $\tilde{Sq}^3$.

Theorem 3.2. *$\mathbb{R}P^2 \times \mathbb{R}P^4$ can be embedded in a complexification X which is a Stein manifold that retracts to $\mathbb{R}P^2 \times \mathbb{R}P^4$.*

3.2.2 Projective Algebraic Manifolds

Theorem 3.3. *Let G be any finite group and n any positive integer ($n > 2$). There exists a projective algebraic manifold X having the same n-type as $K(\mathbb{Z}, 2) \times K(G, 1)$. In particular, $H^*(G;\mathbb{Z})$ is a direct factor of $H^*(X;\mathbb{Z})$ up to dimension n.*

Now let $y \in H^{2q}(X;\mathbb{Z})$ be the cohomology class as in theorem 1.7 and 1.8, then y is the counterexample of a Stein manifold/projective algebraic manifold that is not analytic.